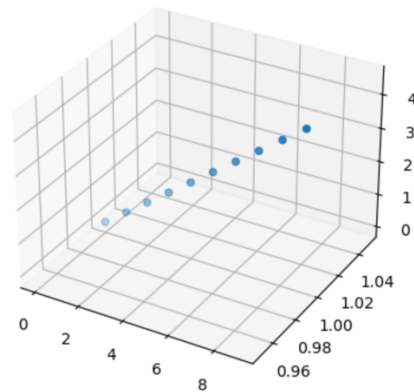
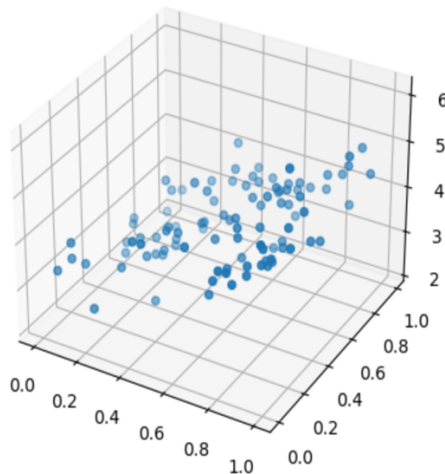


Questions on Linear Regression and Gradient Descent

1. Linear Regression

- Use your closed-form implementation to fit a model to the data in 1D-no-noise-lin.txt and 2D-noisy-lin.txt. Report the loss of the fitted model in both cases. Include plots for both. Note: Ensure that you handle cases where the y-intercept is non-zero.



-
- What happens when you use the closed-form solution and one of the features (columns of X) is duplicated? Explain why. Note: Testing this out with your code may be a useful first step, but you should also critically think about what is happening and why.
 - When one of the features is an exact copy of another feature, then the matrix becomes singular due to linearly dependent nature of the features. As linear regression is under the assumption that features are somewhat independent of one another, when features are duplicated then an error is caused.
- Does the same thing happen if one of the training points (rows of X) is duplicated? Explain why.
 - As this is the training point duplication, this doesn't introduce new information thus is disregarded.
- Does the same thing happen with Gradient Descent? Explain why.
 - It does matter in Gradient Descent as each feature affects the gradient thus a feature that is duplicated has a higher weightage on the gradient algorithm.

2. Gradient Descent

- Now use your Gradient Descent implementation to fit a model to 1D-no-noise-lin.txt with $\alpha=0.05$, $\text{num_iters}=10$, and initial Theta set to the zero vector. Report the output (the full list of (Theta, loss) tuples for each of the 10 iterations).

```

Iteration: 1 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 2 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 3 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 4 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 5 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 6 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 7 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 8 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 9 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 10 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]

```

-
- Using the default parameters for alpha and num_iters, and with initial Theta set to zero, do you get the same model parameters as you did with the closed-form solution? Do you get the same loss? Report this for both 1D-no-noise-lin.txt and 2D-noisy-lin.txt.

```

Iteration: 1 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 2 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 3 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 4 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 5 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 6 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 7 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 8 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 9 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 10 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]

```

2D-noisy-lin.txt

```

Iteration: 1 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 2 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 3 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 4 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 5 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 6 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 7 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 8 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 9 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 10 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]

```

1D-no-noise-lin.txt

-
- As reported the two files both 1D-no-noise-lin.txt and 2D-noisy-lin.txt according to the graphs above depict that with the default parameters the loss and theta are still the same.

- Find a set of values for the learning rate and iterations where you get the same answers and a set of values where the answers are noticeably different. Explain what's going on in both cases, for both datasets 1D-no-noise-lin.txt and 2D-noisy-lin.txt.

<pre> Iteration: 435 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 436 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 437 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 438 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 439 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 440 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 441 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 442 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] Iteration: 443 , Loss: -5.8683245422692805 , Theta: [[4.46715737] [1.97034174]] </pre>	2D-noisy-lin.txt
--	-------------------------

○ Similar-

```

Iteration: 400 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 401 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 402 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 403 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 404 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 405 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 406 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 407 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 408 , Loss: 7.473774949078646 , Theta: [[4.46715737]
[1.97034174]]

```

1D-no-noise-lin.txt

○ Difference-

```

Iteration: 334 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 335 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 336 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 337 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 338 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 339 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 340 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 341 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]
Iteration: 342 , Loss: -5.8683245422692805 , Theta: [[4.46715737]
[1.97034174]]

```

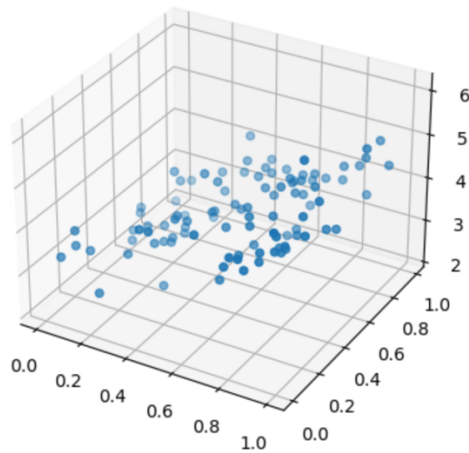
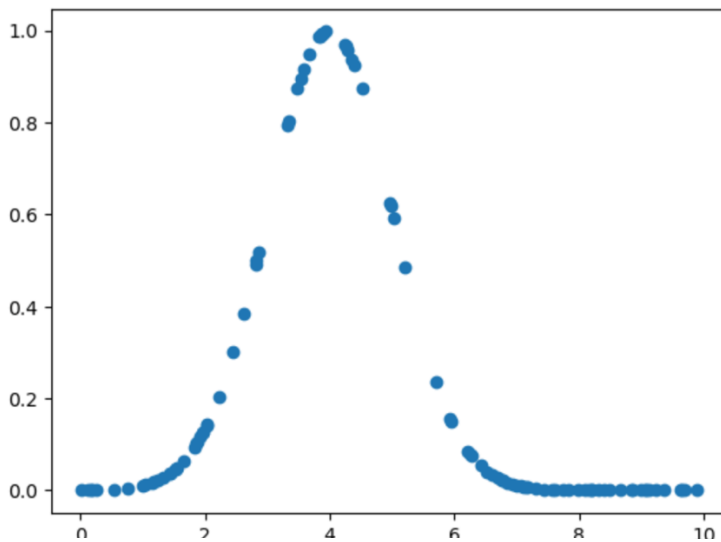
2D-noisy-lin.txt

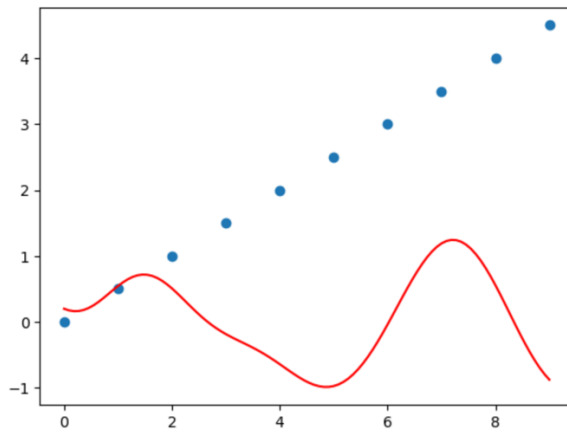
- As seen in the plots above the similarities and differences are portrayed when attempting to get a similarity between the two datasets. This is seen as

convergence is measured with a higher number of iterations and smaller learning rates as this will minimize the cost function. Conversely divergences require for lower number of iterations and higher learning rate (α).

3. Random Fourier Features

- Use your Random Fourier Features implementation to fit models for 1D-exp-samp.txt, 1D-exp-uni.txt, 1D-quad-uni.txt, and 1D-quad-uni-noise.txt, each with 3 different values for K : small, medium, and large. The small value should noticeably under-fit the data, the large value should fit almost perfectly, and the medium value should be somewhere in between. Report the values for K and include graphs for each of the four datasets (so you should report 12 values of K and have 12 graphs in total).





-
- $K = [0, 5, 10, 50, 100, 200, 300, 400, 500, 600, 700, 800]$
- EXTRA CREDIT: What happens with Random Fourier Features if you try to predict a value far away from any points in the training dataset? Modify the plotting functions (or write your own) and show what the model looks like far away from the training points. Describe what you are seeing qualitatively, compare the model's output with the "true" output quantitatively, and explain why this is happening. You may want to run these experiments multiple times to get a better understanding of what is going on.